

String Theory and M-Theory

Lecture 8: Conformal Maps and the Open-String Scattering Prescription

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Companion notes curated by [LazyingArt LLC](#) with [Video2Book](#)

Chapter 1

Conformal Maps and the Open-String Scattering Prescription

Overview. Conformal mapping first appears as a practical symmetry of two-dimensional electrostatics. Complex differentiability turns that symmetry into a calculable one: the Cauchy–Riemann equations imply Laplace’s equation, and a direction-independent complex derivative preserves angles. With this machinery in hand, a string worldsheet may be drawn as a strip, half-plane, cylinder, or disc without changing the physics. Open-string scattering then becomes a Gaussian path integral on the disc with momentum insertions on its boundary, followed by an integral over their inequivalent positions.

1.1 Electrostatics in a Two-Dimensional World

We begin with a literal two-dimensional electrostatic problem. Charges and their field lines are confined to a plane; they are not merely ordinary three-dimensional charges whose motion happens to be planar. Gauss’s law then gives

$$F(r) \propto \frac{1}{r}, \quad V(r) \propto \log r, \quad (1.1)$$

because the flux crosses a circle whose circumference grows like r .

Write the field in terms of a scalar potential ϕ . In the sign convention used on the board,

$$\mathbf{E} = \nabla \phi, \quad \nabla \cdot \mathbf{E} = \rho, \quad (1.2)$$

and therefore

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = \rho. \quad (1.3)$$

In a charge-free region this reduces to Laplace’s equation,

$$\nabla^2 \phi = 0. \quad (1.4)$$

¹Editorial clarification: Many electrodynamics texts define $\mathbf{E} = -\nabla \Phi$, in which case $\nabla^2 \Phi = -\rho$ after constants are absorbed. The lecture uses the opposite potential sign. The source-free equation and all conformal arguments are unchanged.

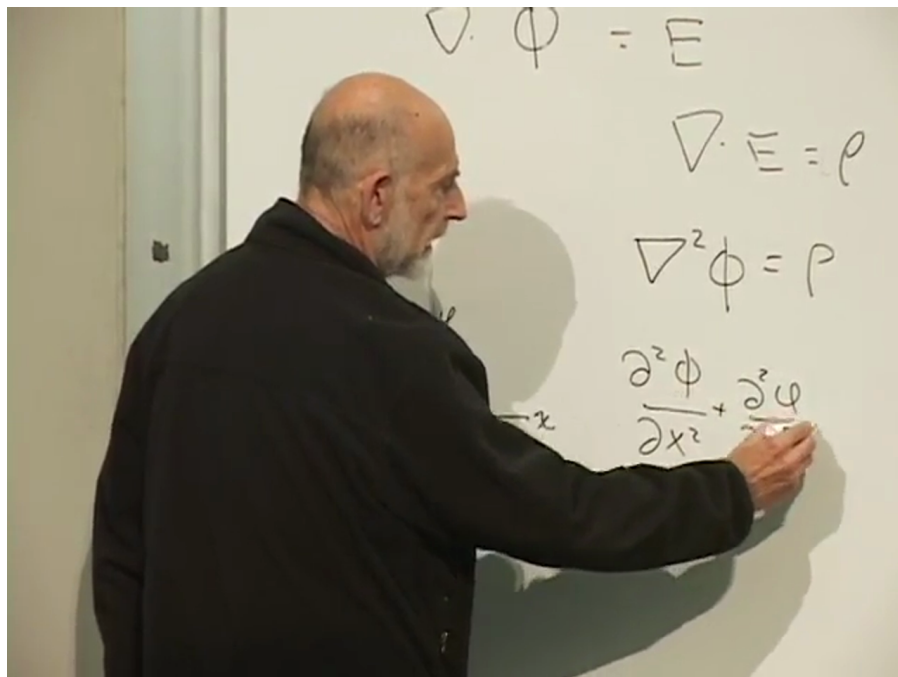


Figure 1.1: The electrostatic equations unfolded into the two-dimensional Poisson operator.

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The plus sign between the two spatial derivatives matters. It makes this an elliptic equation rather than a wave equation. Conformal transformations preserve its source-free form, so a solution on one domain may be carried to a differently shaped domain together with its boundary conditions. This is useful in electrostatics, two-dimensional fluid flow, and cartography before it is useful in string theory.

Question from the audience. If the Laplacian vanishes, is this a standing-wave equation? ^a

Response. No. Both spatial derivatives have the same sign and there is no time derivative. A wave equation requires time dependence with the Lorentzian relative sign. Electrostatics is static precisely because nothing, including a wave, is evolving.

^aLecture timestamp: 00:05:35.

Question from the audience. Is there a curl in two dimensions? ^a

Response. Yes, but it has only one independent component. For $\mathbf{E} = (E_x, E_y)$,

$$\text{curl}_{2d} \mathbf{E} = \partial_y E_x - \partial_x E_y.$$

It may be pictured as the component perpendicular to the board, although an intrinsically two-dimensional world needs no external perpendicular direction. In higher dimensions the corresponding object is naturally an antisymmetric two-form.

^aLecture timestamp: 00:06:06.

1.2 A Derivative with No Preferred Direction

Introduce complex coordinates on the original and image planes:

$$z = x + iy, \quad w = u + iv, \quad w = w(z). \quad (1.5)$$

Here x, y, u, v are real. Locally, the derivative ought to be

$$\frac{dw}{dz} = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z}. \quad (1.6)$$

The complex plane, unlike the real line, permits infinitely many directions of approach. A continuous and ordinarily smooth map need not give the same ratio along each direction.

Compare two simple approaches. Along the x -direction, $dy = 0$, so

$$\left. \frac{dw}{dz} \right|_x = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \quad (1.7)$$

Along the y -direction, $dx = 0$, and

$$\left. \frac{dw}{dz} \right|_y = \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}. \quad (1.8)$$

Equality of real and imaginary parts gives

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}} \quad (1.9)$$

These are the Cauchy–Riemann equations. Under the usual differentiability assumptions they are also sufficient for the limit (1.6) to be direction independent. A function with this property is analytic, or holomorphic, at the point.

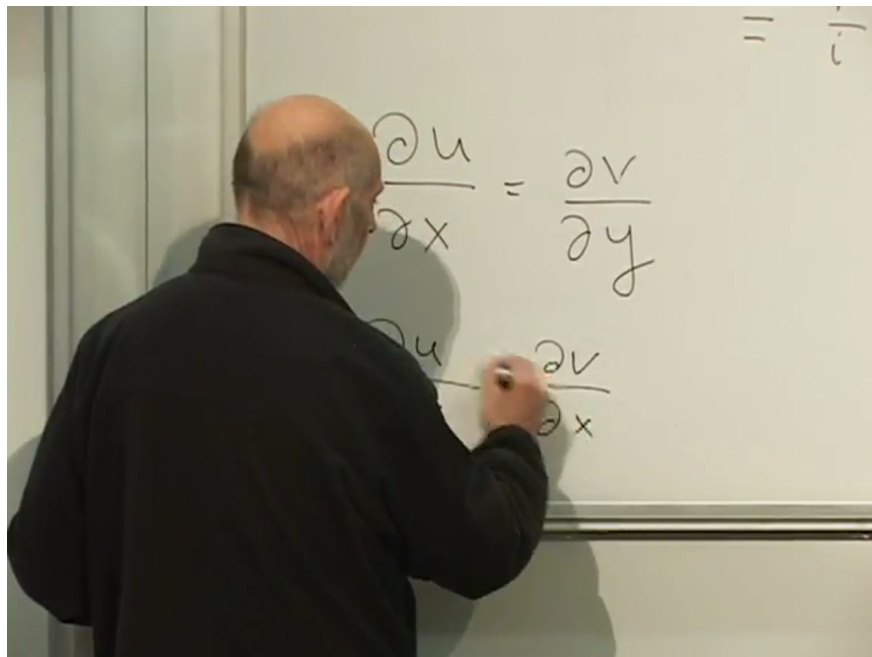


Figure 1.2: The completed Cauchy–Riemann pair, including the relative minus sign.

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The lecture briefly distinguishes related terminology. A function holomorphic throughout the finite complex plane is entire. A meromorphic function may have isolated poles. Thus $1/z$ is analytic away from $z = 0$, but it is not entire.

1.3 Harmonic Conjugates and Conformal Geometry

The Cauchy–Riemann equations are coupled first-order equations, but they imply separate second-order equations. Differentiate $\partial_x u = \partial_y v$ with respect to x , differentiate $\partial_y u = -\partial_x v$ with respect to y , and add:

$$\partial_x^2 u = \partial_x \partial_y v, \quad (1.10)$$

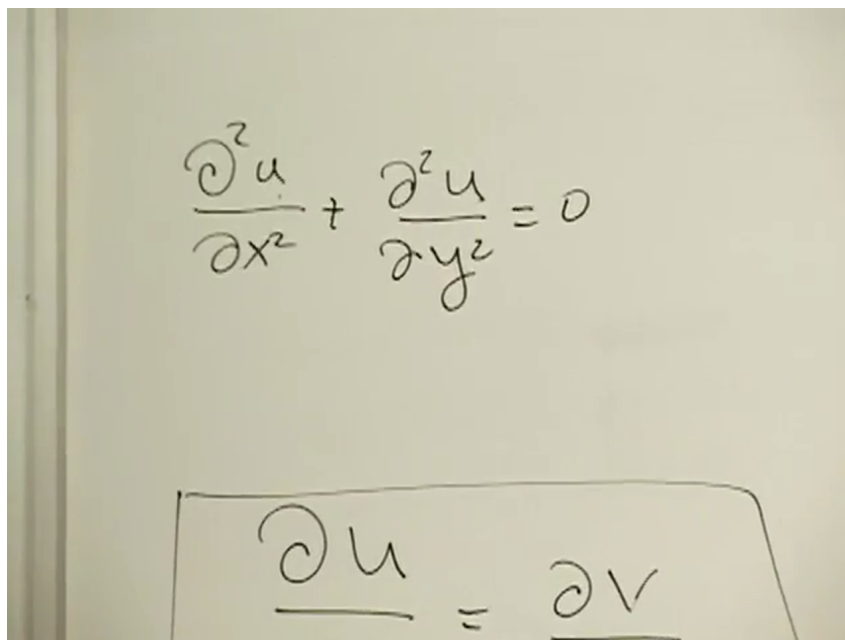
$$\partial_y^2 u = -\partial_y \partial_x v, \quad (1.11)$$

$$\therefore \quad \partial_x^2 u + \partial_y^2 u = 0. \quad (1.12)$$

Repeating the argument with u and v interchanged gives

$$\partial_x^2 v + \partial_y^2 v = 0. \quad (1.13)$$

The real and imaginary parts of an analytic function are therefore harmonic conjugates. Two arbitrary harmonic functions do not automatically make an analytic function; they must still be linked by (1.9).



$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}}$$

Figure 1.3: The mixed derivatives cancel, leaving Laplace’s equation for one component of the analytic map.

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There is also a useful vector interpretation. If $\mathbf{F} = (u, -v)$, then the two Cauchy–Riemann equations state, up to the chosen orientation convention, that both the divergence and the two-dimensional curl of $\mathbf{F}$ vanish. This is the promised return to electrostatics.

1.3.1 Why Angles Are Preserved

Write two small complex displacements in polar form:

$$\delta z = \rho e^{i\theta}, \quad \Delta z = \rho' e^{i\theta'}. \quad (1.14)$$

Their ratio has phase $\theta - \theta'$, exactly the angle between them. Under an analytic map,

$$\frac{\delta w}{\delta z} = \frac{\Delta w}{\Delta z} = w'(z), \quad (1.15)$$

because the derivative does not depend on direction. Rearranging gives

$$\frac{\Delta z}{\delta z} = \frac{\Delta w}{\delta w}. \quad (1.16)$$

The two sides have the same phase, so the angle between the original intervals equals the angle between their images.

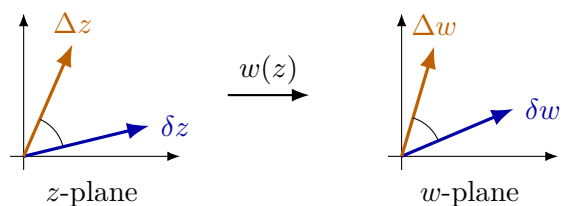
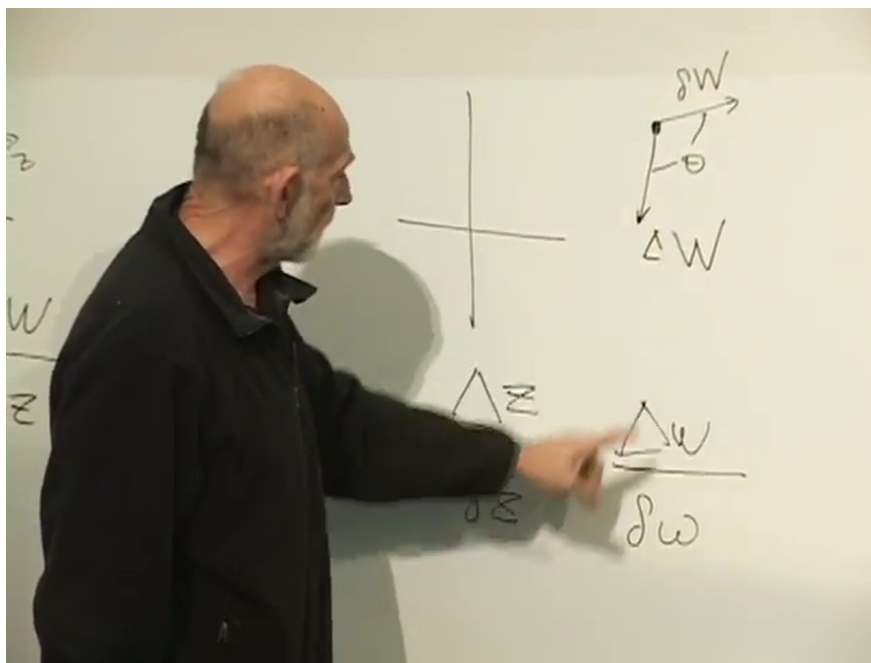


Figure 1.4: The board proof and a clean local reconstruction. Multiplication by $w'(z)$ rotates and rescales every infinitesimal direction by the same amount, preserving their relative angle.

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This local argument assumes $w'(z) \neq 0$. At a critical point the map may fail to be locally one-to-one, and one must treat the geometry separately. Away from such points, analytic maps and orientation-preserving conformal maps are locally the same objects.

1.4 Examples that Train the Eye

1.4.1 Squaring and Complex Conjugation

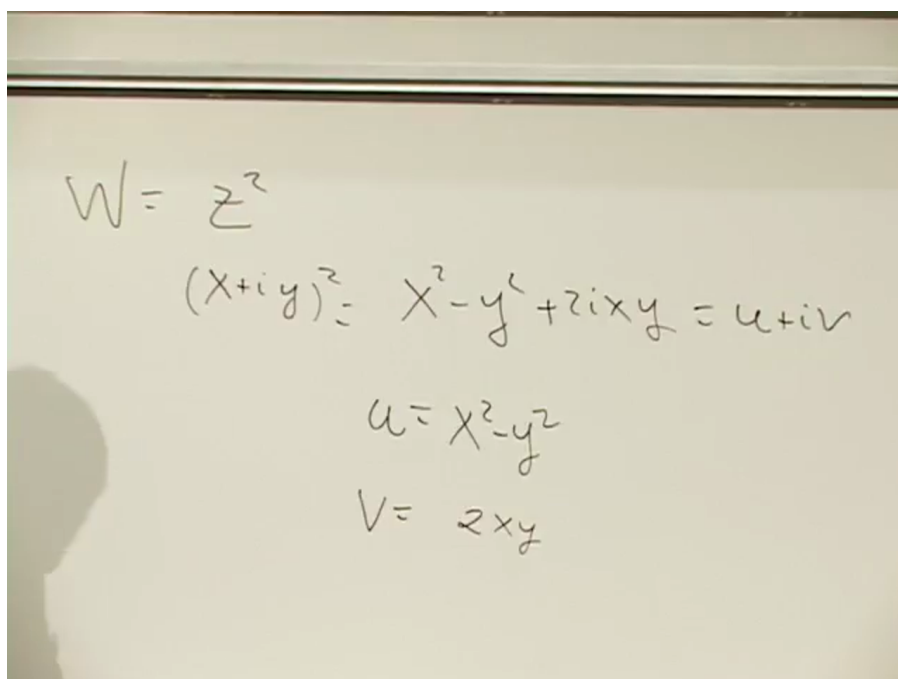
For $w = z^2$,

$$(x + iy)^2 = x^2 - y^2 + 2ixy = u + iv, \quad (1.17)$$

so

$$u = x^2 - y^2, \quad v = 2xy. \quad (1.18)$$

The second derivatives of u are 2 and -2 ; those of v vanish separately. Both are harmonic, and direct differentiation also checks the Cauchy–Riemann equations.



Handwritten notes on a whiteboard:

$$w = z^2$$

$$(x + iy)^2 = x^2 - y^2 + 2ixy = u + iv$$

$$u = x^2 - y^2$$

$$v = 2xy$$

Figure 1.5: The squaring map separated into its real and imaginary harmonic components.

Lecture frame: 00:36:45.

Question from the audience. Does $1/z$ give a nonanalytic example? ^a

Response. Only at one point. It is analytic wherever $z \neq 0$ and has a pole at the origin. To see a map that fails the Cauchy–Riemann test everywhere, complex conjugation is a better example.

^aLecture timestamp: 00:37:49.

By contrast,

$$w = \bar{z} = x - iy \quad \implies \quad u = x, \quad v = -y \quad (1.19)$$

reverses orientation. Its derivatives fail the first Cauchy–Riemann equation by a sign, so it is not analytic as a function of z .

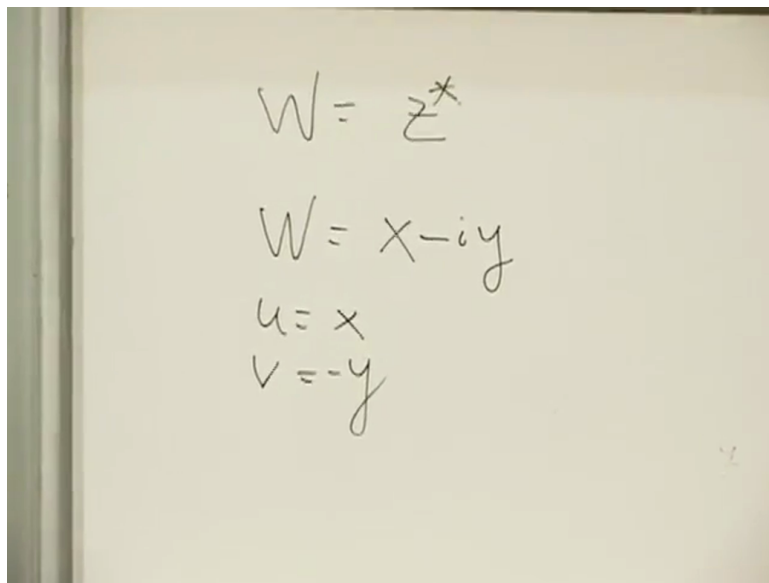


Figure 1.6: Complex conjugation provides the deliberately nonanalytic counterexample.

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1.4.2 The Exponential and Its Inverse

Question from the audience. If e^z is analytic, why should its inverse $\log z$ also be analytic?
^a

Response. Where the inverse exists and $w'(z) \neq 0$, its derivative is $1/w'(z)$. Direction independence is therefore preserved. Globally the logarithm is multivalued, so one chooses a branch and excludes its branch point and cut. The local conformal statement remains valid on that chosen domain.

^aLecture timestamp: 00:40:38.

Euler's formula gives

$$w = e^z = e^{x+iy} = e^x(\cos y + i \sin y), \quad (1.20)$$

hence

$$u = e^x \cos y, \quad v = e^x \sin y. \quad (1.21)$$

For example, $\partial_x u = e^x \cos y = \partial_y v$, and the second Cauchy–Riemann equation follows just as directly.

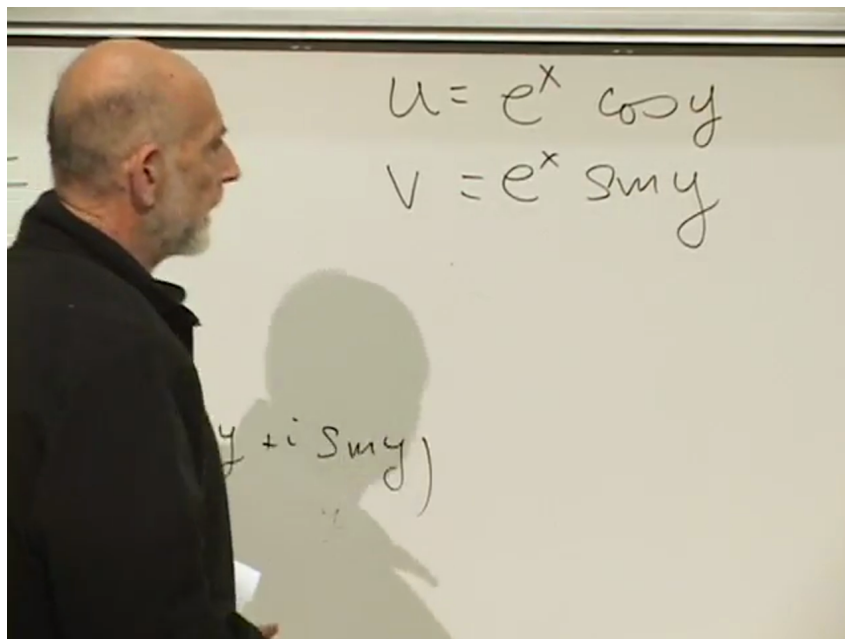


Figure 1.7: The real and imaginary components of the exponential map.

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1.5 Half-Plane, Strip, Cylinder, and Disc

1.5.1 The Logarithm Turns Radius into Time

Take the right half-plane, where $-\pi/2 < \theta < \pi/2$, and write

$$z = re^{i\theta}, \quad w = \log z = \log r + i\theta. \quad (1.22)$$

The half-plane maps to the infinite strip

$$-\frac{\pi}{2} < \operatorname{Im} w < \frac{\pi}{2}. \quad (1.23)$$

Rays of fixed θ become horizontal lines, while semicircles of fixed r become vertical lines. The origin is sent to $\operatorname{Re} w = -\infty$, and $r = \infty$ to $\operatorname{Re} w = +\infty$.

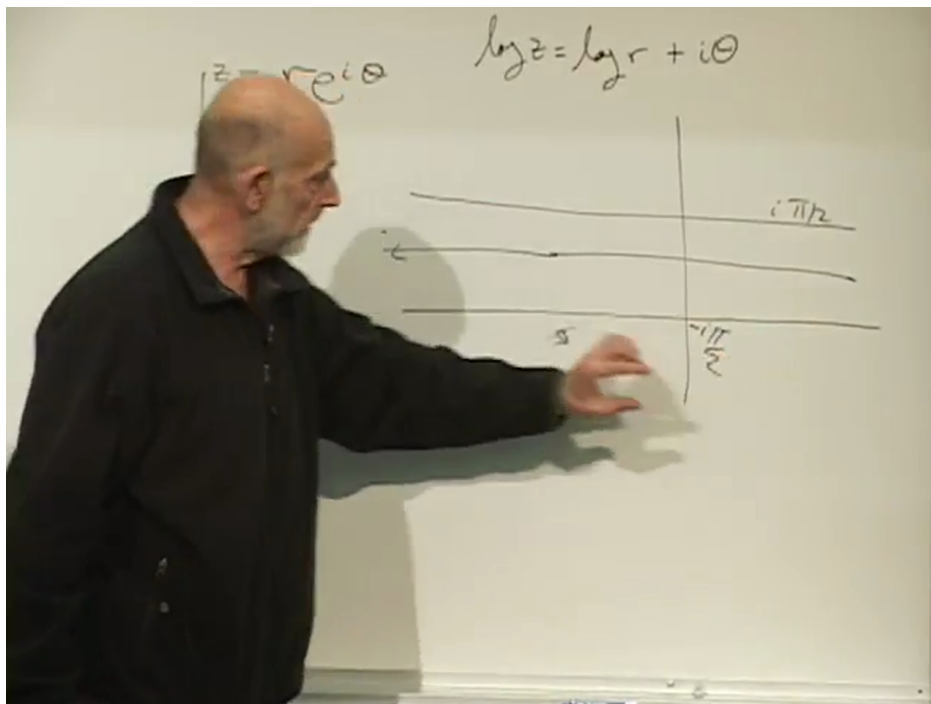


Figure 1.8: The logarithm carries the right half-plane into an infinite strip: polar rays and semicircles become an orthogonal rectangular grid.

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The strip is the standard worldsheet of an open string. Its bounded coordinate is the parameter along the string, while translation along the unbounded direction is worldsheet time. Under the inverse exponential map, a time translation becomes a dilation $z \mapsto e^a z$. Thus the same symmetry looks like translation on the strip and uniform expansion on the half-plane.

Question from the audience. Where does a particular value such as $r = 1$ go, and are the coordinates on the strip still polar? ^a

Response. A circle of fixed radius becomes a vertical line at $\operatorname{Re} w = \log r$; in particular $r = 1$ maps to the central line $\operatorname{Re} w = 0$. Polar coordinates describe the z -plane. The w -plane uses ordinary rectangular coordinates $(\log r, \theta)$.

^aLecture timestamp: 00:50:58.

If the full punctured plane is mapped by $\log z$, the angular coordinate has period 2π . The resulting strip therefore has its upper and lower edges identified: it is an infinite cylinder. The half-plane with a boundary describes the open-string worldsheet; the full plane, with no boundary and periodic angle, describes a closed-string cylinder.

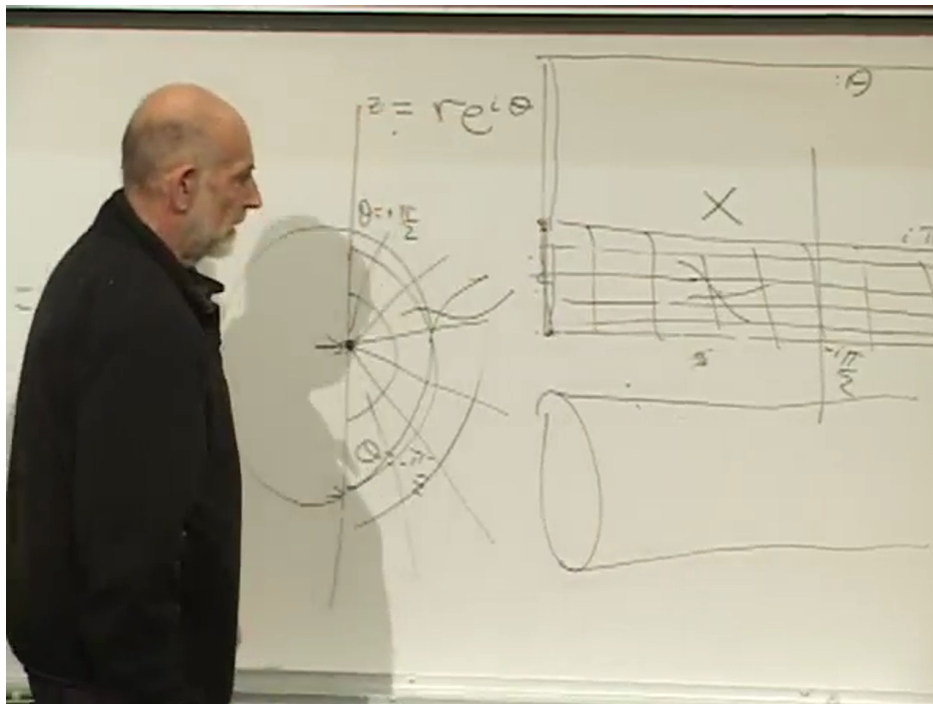


Figure 1.9: The half-plane/strip picture and the periodic identification that turns the full punctured plane into a closed-string cylinder.

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1.5.2 A Cayley Map to the Unit Disc

A linear fractional, or Möbius, map carries the right half-plane to the unit disc:

$$w = \frac{z - 1}{z + 1}. \quad (1.24)$$

It sends $z = 0$ to -1 , $z = \infty$ to $+1$, and the imaginary axis to $|w| = 1$. If $z = ir$,

$$\left| \frac{ir - 1}{ir + 1} \right| = 1. \quad (1.25)$$

Points with $\text{Re } z > 0$ lie inside the circle; the left half-plane maps outside it. ²

²Editorial clarification: During the live derivation the reciprocal $(z + 1)/(z - 1)$ is written while the interior-disc image is described. That reciprocal has the same unit-circle boundary but sends the right half-plane to the exterior. Equation (1.24) is the interior-disc convention consistent with the stated geometry.

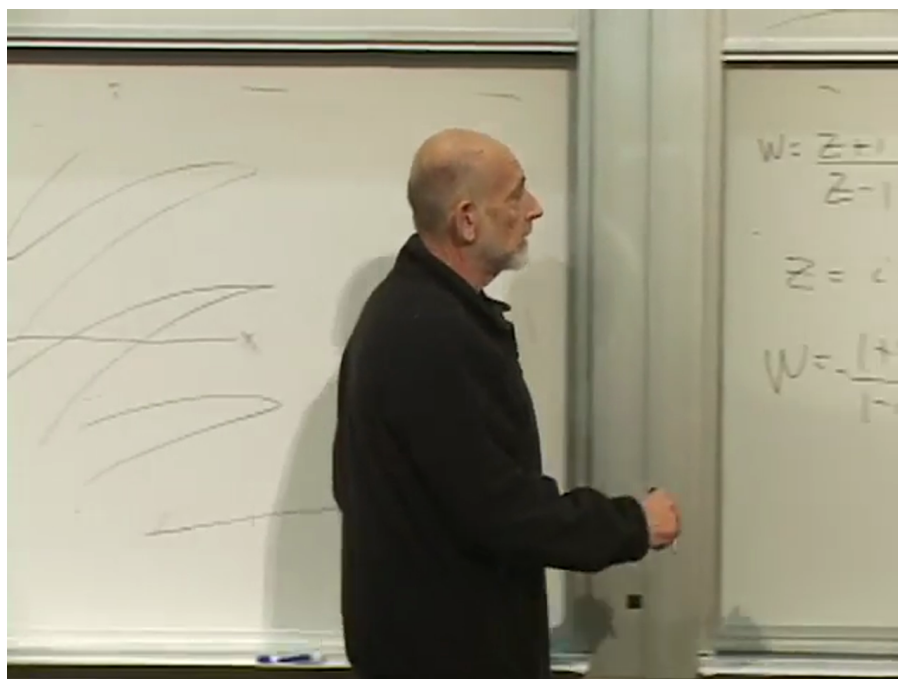


Figure 1.10: The linear-fractional construction of the unit-circle boundary. The reciprocal convention visible on the board exchanges interior and exterior.

Lecture frame: 00:58:38.

Möbius maps send the generalized collection of circles and straight lines to circles and straight lines. The radial and circular grid on the half-plane consequently becomes two orthogonal families of arcs on the disc. A straight diameter is simply the limiting case of a circle whose center has gone to infinity.

Question from the audience. Are the arcs drawn inside the disc really circles? ^a

Response. Yes. Their centers occur at different locations, often outside the visible disc. The diameter is the infinite-radius limit of the same family.

^aLecture timestamp: 01:00:55.

Question from the audience. Does the disc drawing mean that it is a sphere, and where does the other half-plane go? ^a

Response. The drawing here is a flat disc, not a sphere. The right half-plane maps inside its boundary and the left half-plane maps outside. If one adds the point at infinity to the entire complex plane, that extended plane can be represented as a sphere by stereographic projection; that is a different global statement.

^aLecture timestamp: 01:01:21.

Question from the audience. Which disc curves correspond to the original radial spokes and constant-radius semicircles? ^a

Response. The positive real axis becomes the diameter joining -1 and $+1$. Other rays become circular arcs joining those same boundary points. Semicircles of fixed radius become the orthogonal family that begins and ends on the unit circle.

^aLecture timestamp: 01:02:59.

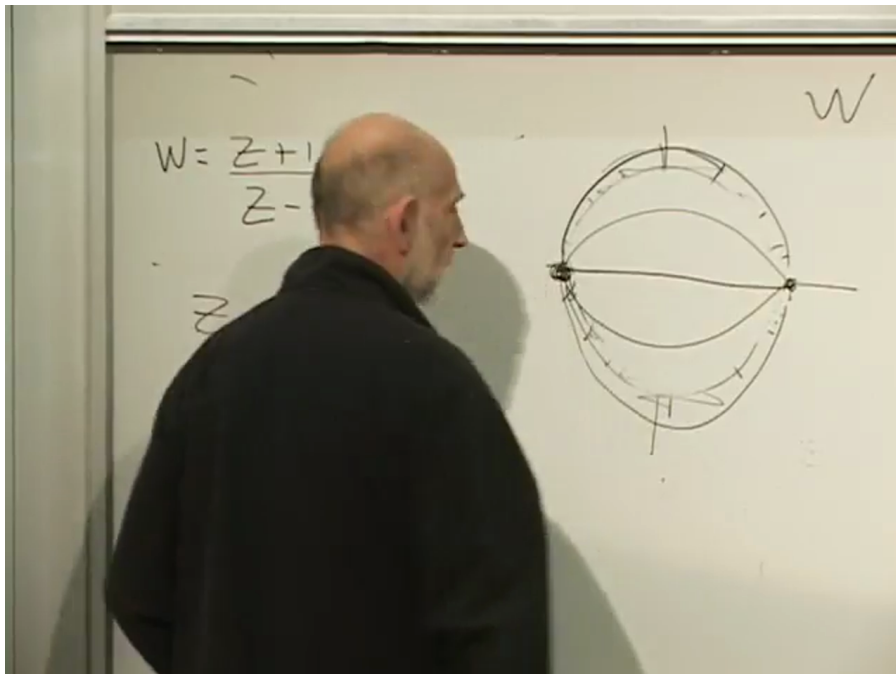


Figure 1.11: Lines and semicircles in the half-plane become orthogonal circle-and-line families in the disc.

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1.6 The Four-String Worldsheet

We can now return to scattering. Two open strings enter, join into one string, propagate for an arbitrary interval, and split into two outgoing strings. In strip coordinates the worldsheet looks like a long ribbon slit from each end—the "sports bandage" picture. The intermediate lifetime is not chosen in advance. It is one of the histories over which the amplitude is integrated.

Let (σ, τ) be coordinates on the worldsheet and let

$$X^\mu = X^\mu(\sigma, \tau) \tag{1.26}$$

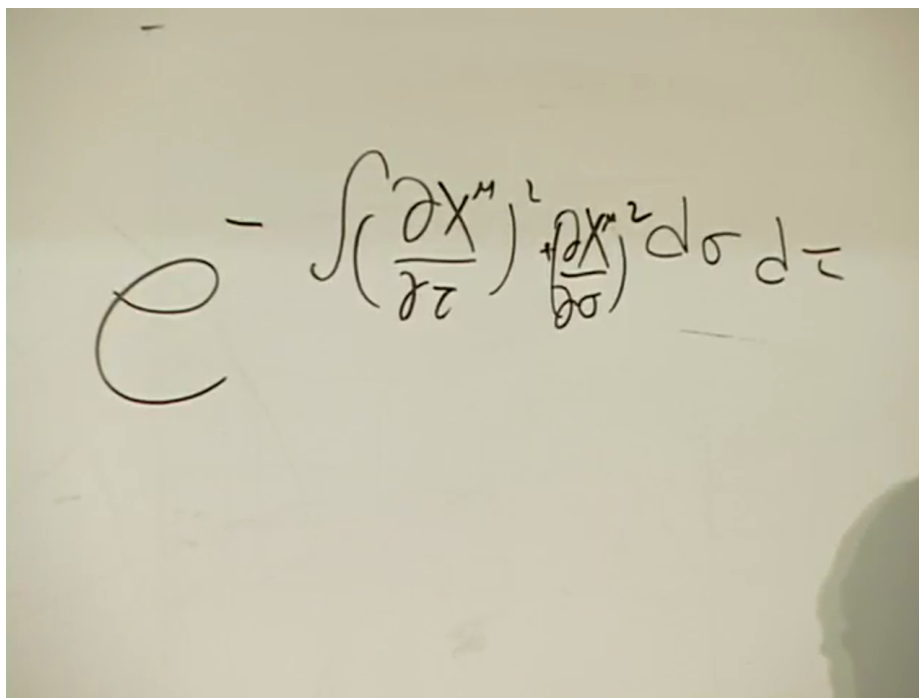
give its position in spacetime. The labels σ, τ live on the two-dimensional parameter surface; the index μ labels spacetime coordinates. In the bosonic theory $\mu = 0, \dots, 25$; in the superstring case there are ten spacetime coordinates.

After Wick rotation, the free embedding fields have Euclidean action

$$S_E[X] = \frac{1}{4\pi\alpha'} \int_\Sigma d\sigma d\tau [\partial_\tau X^\mu \partial_\tau X_\mu + \partial_\sigma X^\mu \partial_\sigma X_\mu]. \tag{1.27}$$

The corresponding path integral sums over every embedding of the chosen worldsheet into spacetime:

$$\int \mathcal{D}X e^{-S_E[X]}. \quad (1.28)$$



A photograph of a whiteboard with a handwritten equation in black marker. The equation is
$$e^{-\int \left(\left(\frac{\partial X^\mu}{\partial \tau} \right)^2 + \left(\frac{\partial X^\mu}{\partial \sigma} \right)^2 \right) d\sigma d\tau}$$
 The handwriting is cursive and slightly slanted. The background is a plain whiteboard surface.

Figure 1.12: The completed positive Euclidean worldsheet weight. The two derivative terms are summed over all spacetime components.

Lecture frame: 01:11:34.

The slit strip has one connected boundary. Conformal freedom therefore allows it to be represented as a disc. Four remote ends of the strips become four marked points on the boundary. Deforming the marked points changes the apparent separation between joining and splitting in strip coordinates, but it does not change the underlying surface or physics.

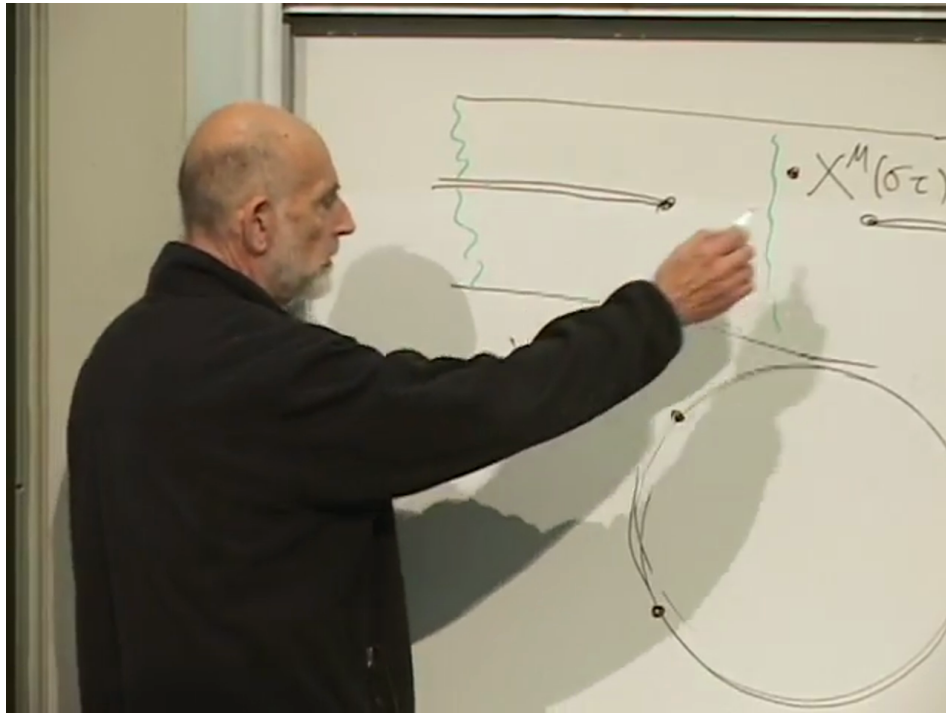


Figure 1.13: The slit-strip worldsheet and its disc representation with boundary points for the external strings.

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The detailed grid lines used to picture this conformal map carry no physical content. What matters is the topology, the boundary, and the marked external points. The same worldsheet can be stretched into different channel pictures without introducing a new interaction.

1.7 External States as Boundary Insertions

The bare integral (1.28) does not yet say which momenta enter and leave. For a scalar open-string state of momentum k_i^μ , insert on the boundary the vertex operator

$$V_i(z_i) = e^{ik_i \cdot X(z_i)}, \quad k_i \cdot X = k_{i\mu} X^\mu. \quad (1.29)$$

The point z_i is a position on the worldsheet boundary. It is not a spacetime coordinate and not a direction of momentum.

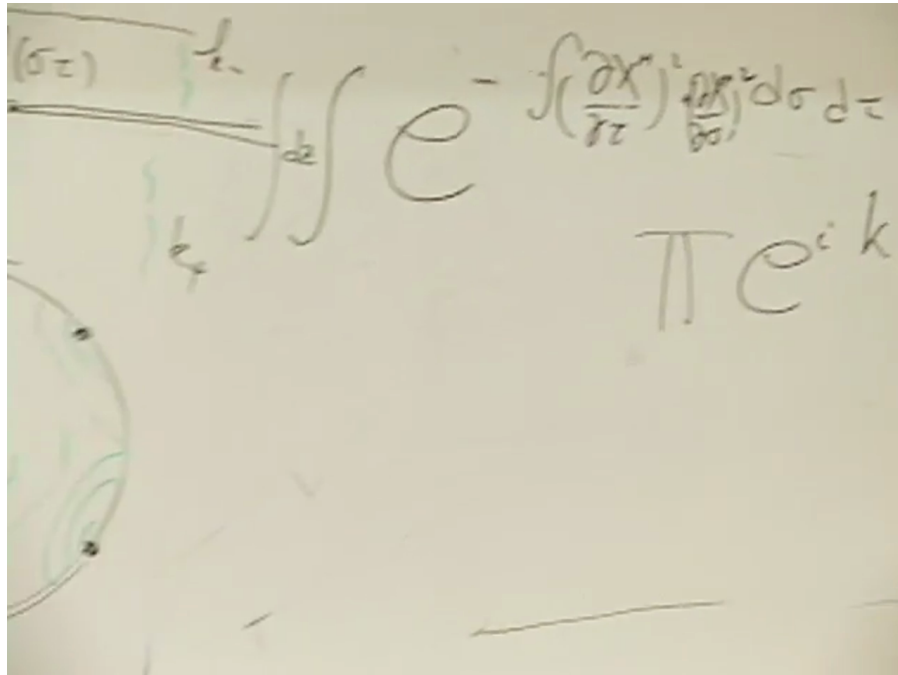


Figure 1.14: The bulk Euclidean weight and the product of boundary momentum insertions are displayed as distinct factors.

Lecture frame: 01:18:58.

For n external open strings, the schematic prescription is

$$\mathcal{A}_n(k_1, \dots, k_n) = \int_{\mathcal{M}_{0,n}^\partial} d\mu(z_i) \int \mathcal{D}X e^{-S_E[X]} \prod_{i=1}^n e^{ik_i \cdot X(z_i)}. \quad (1.30)$$

The functional integral is over embeddings X^μ . The finite-dimensional integral is over inequivalent boundary insertion positions. Polarization factors and the gauge-fixing measure are suppressed at this stage, as they are in the lecture. ³

Question from the audience. If the external momenta are fixed, why are they integrated over? ^a

Response. They are not. The k_i are fixed labels of the scattering amplitude. What is integrated are the embedding fields and the worldsheet locations z_i at which those fixed momenta are inserted. The latter integral replaces the integral over the intermediate lifetime in the strip picture.

^aLecture timestamp: 01:20:42.

³Editorial clarification: A complete gauge-fixed string amplitude also contains the appropriate ghost contribution, state-dependent vertex factors, coupling constants, and normalization. Equation (1.30) records the embedding-field and position dependence developed in this lecture.

Question from the audience. Is the product of vertex factors inside the path integral, and is it part of the action? ^a

Response. It multiplies e^{-S_E} inside the functional integral. One may equivalently exponentiate the factors and regard them as localized source terms added to the action, but they are conceptually vertex operators rather than part of the free bulk action.

^aLecture timestamp: 01:21:39.

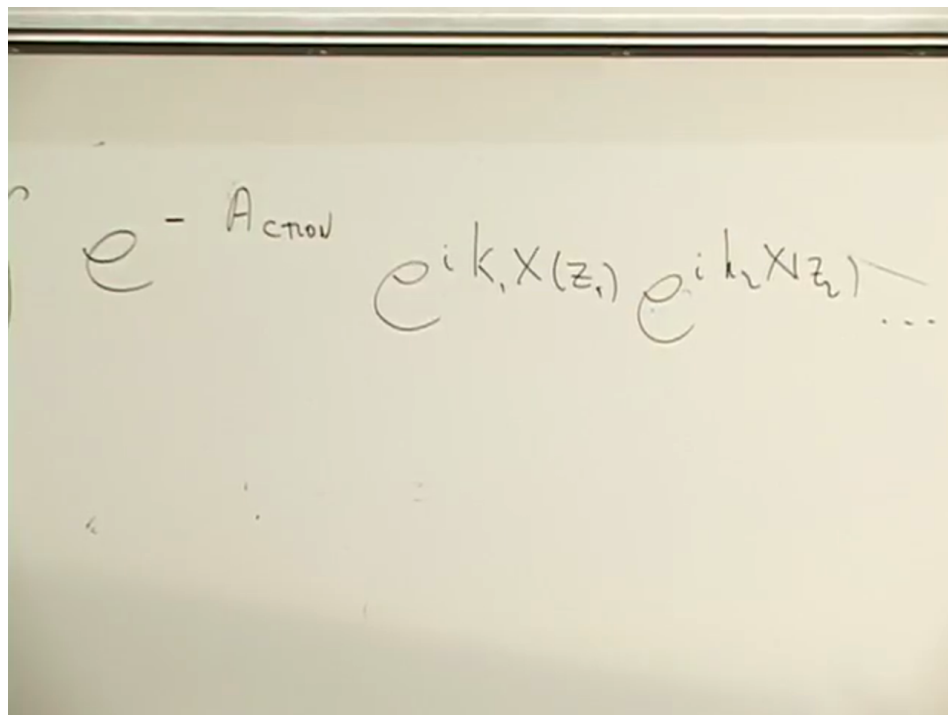


Figure 1.15: The path integral written with separate factors $e^{ik_i \cdot X(z_i)}$ for successive external particles.

Lecture frame: 01:23:23.

Question from the audience. Why does $e^{ik \cdot X}$ still contain i after the worldsheet Wick rotation? ^a

Response. The rotation changes the worldsheet time and converts the bulk weight from e^{iS} to e^{-S_E} . The i in a momentum eigenfunction is the Fourier phase that labels spacetime momentum. It has a different role and remains.

^aLecture timestamp: 01:23:42.

Question from the audience. Does the position z_i determine the direction of the momentum vector? ^a

Response. No. Direction is encoded in the spacetime components k_i^μ . The insertion point z_i says where that momentum enters the worldsheet description; $X^\mu(z_i)$ is among the field values summed over by $\mathcal{D}X$.

^aLecture timestamp: 01:24:19.

Question from the audience. Do the drawn angles of the bandage legs encode the directions of the physical momenta? ^a

Response. No. The bandage is parameter space, not an image of the string in physical spacetime. Its legs may be drawn horizontally, tilted, or conformally bent. Physical directions are carried by the components k^μ , while $X^\mu(\sigma, \tau)$ is integrated over.

^aLecture timestamp: 01:25:07.

1.8 The Gaussian Integral as Electrostatics

The action (1.27) is quadratic in every component X^μ , and the insertions are linear sources once their exponentials are combined with the action. The functional integral is therefore Gaussian. At its saddle point, each spacetime component obeys a Poisson equation of the schematic form

$$\nabla^2 X^\mu(z) \propto i\alpha' \sum_i k_i^\mu \delta(z - z_i), \quad (1.31)$$

with the open-string boundary condition on the rim of the disc. Each X^μ behaves like an independent electrostatic potential, and each momentum component k_i^μ behaves like the corresponding charge placed at z_i .

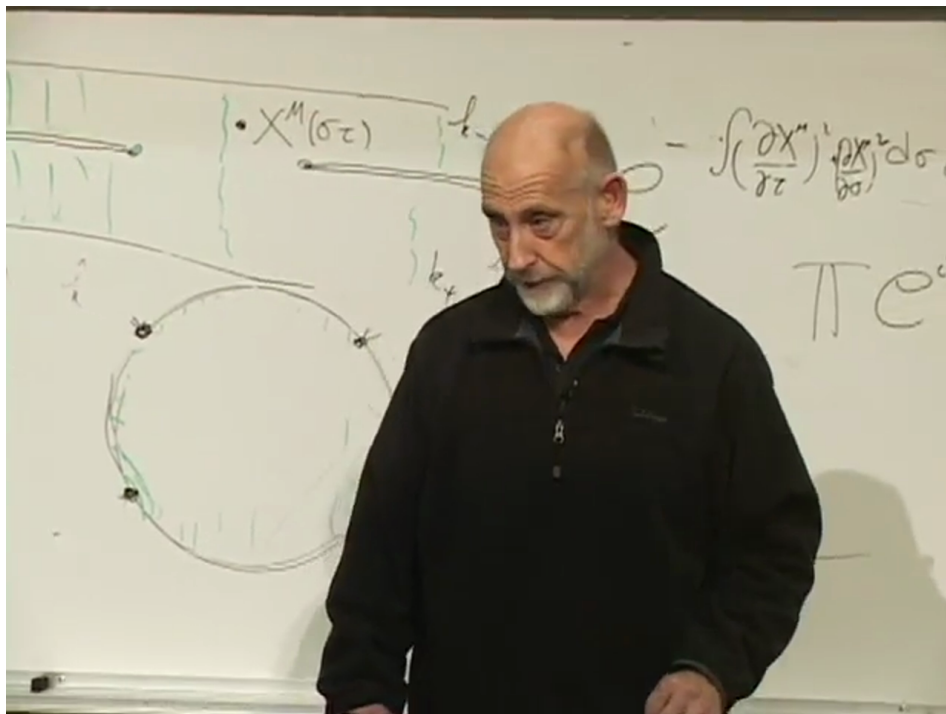


Figure 1.16: The disc with boundary sources is interpreted as one independent two-dimensional electrostatic problem for each spacetime component.

Lecture frame: 01:28:58.

Solve the electrostatic problem, evaluate its energy, exponentiate the negative energy, and finally integrate over the inequivalent charge positions. In standard open-string normalization, the

embedding-field correlator supplies the Koba–Nielsen factor

$$\left\langle \prod_i e^{ik_i \cdot X(z_i)} \right\rangle \propto (2\pi)^D \delta^{(D)} \left(\sum_i k_i \right) \prod_{i < j} |z_i - z_j|^{2\alpha' k_i \cdot k_j}. \quad (1.32)$$

The delta function expresses momentum conservation. The product contains the electrostatic pair energies. For the four scalar boundary insertions used here, fixing three positions turns the remaining integral into the Euler beta integral encountered in the preceding lectures.

Question from the audience. Are there twenty-six electrostatic energies or only one? ^a

Response. There is one independent potential and field for each spacetime component. Their quadratic energies have no cross terms in an orthonormal target-space basis, and the contributions are added to one total action before exponentiation.

^aLecture timestamp: 01:30:53.

Question from the audience. Is the angular coordinate of the disc the physical scattering angle? ^a

Response. No. It is a worldsheet position parameter. Its inequivalent combination is related to the intermediate propagation interval and is integrated. The physical scattering angle is encoded through Lorentz invariants built from the fixed momenta.

^aLecture timestamp: 01:32:10.

Question from the audience. Why does this construction fail away from the critical dimension? ^a

Response. The formal Gaussian integrals alone are not the whole consistency condition. Gauge fixing introduces a conformal anomaly whose matter and ghost contributions cancel only at the critical dimension: $D = 26$ for the bosonic string and $D = 10$ for the superstring. Without that cancellation the supposed conformal symmetry of the quantum worldsheet is lost.

^aLecture timestamp: 01:32:35.

Question from the audience. Have we already assumed three large spatial dimensions and compact extra dimensions? ^a

Response. No. Compactification has not yet entered the construction. At this stage the bosonic coordinates still range over all twenty-six target-space directions, or ten in the superstring case. How a subset can remain macroscopically large is the subject of the next discussion.

^aLecture timestamp: 01:33:38.

Question from the audience. Is the connection with electrostatics simply that the coordinates satisfy Laplace’s equation? ^a

Response. Exactly. Away from the insertion points each X^μ is harmonic; the vertex operators supply delta-function sources. That is mathematically the same boundary-value problem as a static two-dimensional potential with point charges, whether or not we use the electrostatic vocabulary.

^aLecture timestamp: 01:33:57.

1.8.1 Conformal Redundancy and the Number of Moduli

The disc has a three-real-parameter conformal automorphism group, $\text{PSL}(2, \mathbb{R})$. Three ordered boundary points may therefore be fixed at convenient locations. For four external strings only the fourth position remains to be integrated; for n ordered boundary insertions there are

$$n - 3 \quad (1.33)$$

real position integrals at tree level.

Question from the audience. Are there four independent angular positions around the disc?

^a

Response. Not physically. One initially writes four positions, but disc conformal symmetry can place any three wherever convenient. Only the relative location of the fourth remains for the two-in/two-out amplitude.

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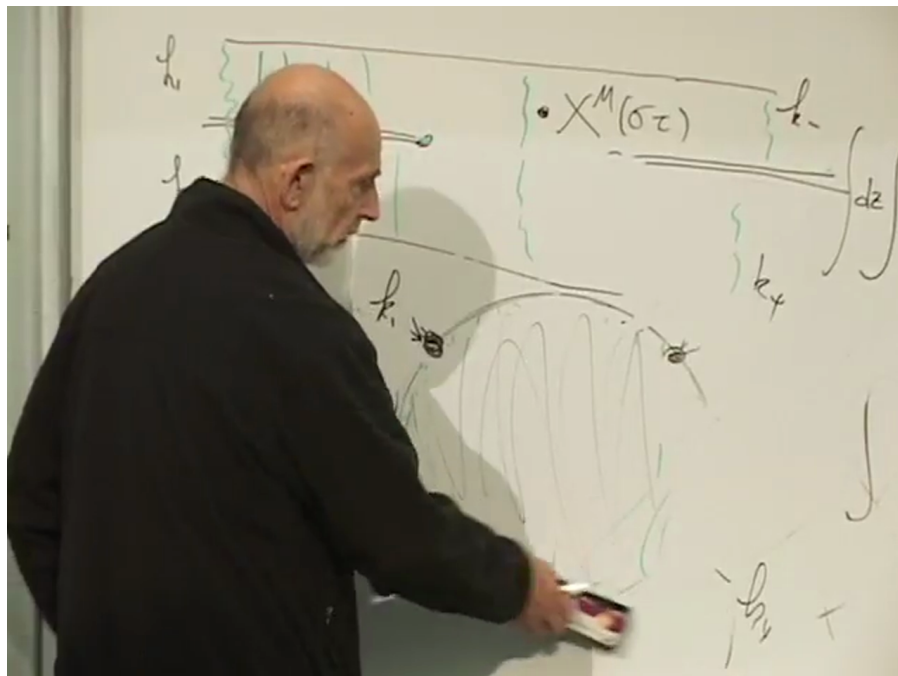


Figure 1.17: Four boundary insertions on the disc. Conformal freedom fixes any three, leaving one four-point modulus.

Lecture frame: 01:35:18.

Question from the audience. What if the two split points do not occur at the same height in the strip drawing? ^a

Response. A conformal map can move that apparent mismatch away. Different drawings are coordinate representatives of the same modulus. The invariant integration is the one over the remaining boundary position, equivalently over the propagation interval.

^aLecture timestamp: 01:35:24.

Question from the audience. For more external particles, how many position parameters remain? ^a

Response. Fix three insertion points with conformal symmetry and integrate the other $n - 3$. This is the boundary-moduli count for an ordered tree-level open-string amplitude.

^aLecture timestamp: 01:36:00.

Question from the audience. How is this related to M-theory? ^a

Response. Not yet. The construction here is perturbative open-string scattering on a worldsheet. Its relation to M-theory requires later duality and nonperturbative ideas, so the question is explicitly deferred.

^aLecture timestamp: 01:36:37.

Question from the audience. Is a different vertical or crossed strip picture a different calculation? ^a

Response. No. One may hold a convenient point fixed and let the remaining modulus range through its full interval. The strip pictures and the boundary-point integral on the disc are different slicings of the same integral, which is why all external strings can be treated symmetrically.

^aLecture timestamp: 01:36:49.

1.9 What the Prescription Does Not Yet Solve

The disc formula extends immediately to more external strings: add more vertex operators and integrate their $n - 3$ inequivalent boundary positions. It is a precise tree-level prescription for a consistent string model, but it is not yet a realistic model of observed particle physics. The simple bosonic theory has the wrong spacetime dimension and additional physical defects. Superstrings improve the spectrum and lower the critical dimension to ten, but six spatial dimensions still have to be compactified.

Question from the audience. Does this simple model already agree with particle collision experiments? ^a

Response. No. The resonance structure historically resembles hadronic scattering, but the twenty-six-dimensional bosonic model cannot be a direct description of observed particles. The formalism defines scattering amplitudes for the model; matching nature requires a realistic spectrum, compactification, interactions, and vacuum choice.

^aLecture timestamp: 01:39:06.

Question from the audience. Will compactification reduce the extra dimensions and produce realistic particle data? ^a

Response. It can reduce the effective macroscopic dimension and generate particle spectra and gauge groups. It is also where much of the real mathematical difficulty enters. Many consistent compactifications are known, and selecting a unique model of observed elementary particles remains unresolved.

^aLecture timestamp: 01:41:10.

The lecture emphasizes this distinction rather than advertising a finished theory. String constructions are mathematically definite and have produced deep information about gravity and geometry, yet their multiplicity is itself part of the physical problem.

Question from the audience. Is there a short way to remember why the bosonic critical dimension is twenty-six? ^a

Response. The earlier oscillator calculation used the regularized zero-point sum $1+2+3+\cdots = -1/12$. The transverse oscillator contribution involves twenty-four directions, and restoring the two longitudinal spacetime coordinates gives $24+2=26$. This is a mnemonic for the earlier consistency argument, not a new derivation here.

^aLecture timestamp: 01:43:19.

Question from the audience. Are the inserted momenta k_i themselves quantized? ^a

Response. In noncompact spacetime their components are continuous. What is quantized is the oscillator level and hence the allowed mass-squared spectrum. Compact dimensions may quantize the corresponding momentum and winding components, but that belongs to the next discussion.

^aLecture timestamp: 01:43:46.